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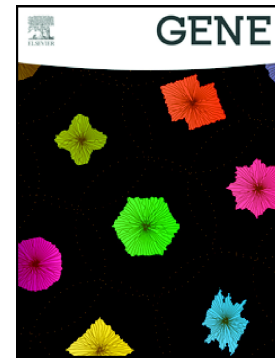
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Differential BDNF methylation in combat exposed veterans the association with exercise

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**Differential *BDNF* methylation in combat exposed veterans the association with exercise****Running Title: *BDNF* methylation in veterans****Joanne Voisey <sup>a\*</sup>; Bruce Lawford <sup>a</sup>; Dagmar Bruenig <sup>b,c</sup>; Wendy Harvey <sup>c</sup>; Charles P Morris <sup>a</sup>; Ross McD Young <sup>b</sup>; Divya Mehta <sup>b</sup>**

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**Abstract**

Brain-derived neurotrophic factor (BDNF) gene is associated with increased risk of posttraumatic stress disorder (PTSD) and plays a role in neuroplasticity, cognition and memory. BDNF has strong potential as a therapeutic target as studies have shown that antidepressants, electroconvulsive treatment and exercise modulate *BDNF* expression and methylation. In this study we examined the role of *BDNF* methylation and expression in PTSD and the implications of exercise in mediating these effects.

*BDNF* DNA methylation and gene expression analysis was performed in a sample of 96 male Vietnam veterans. Cases were combat-exposed veterans with current PTSD ( $n = 48$ ) and controls were combat exposed veterans with no past or current PTSD diagnosis ( $n = 48$ ).

No association between *BDNF* mRNA and PTSD was identified. PTSD was associated with decreased methylation at three *BDNF* CpG sites (cg01546433  $P = 0.004835$ ; cg24650785  $P = 0.000259$  and cg002298481  $P = 0.000672$ ). Differential *BDNF* methylation was associated with exercise, with active exercise associated with lower methylation levels at three CpG sites (cg04481212  $P = 0.005$ ; cg01546433  $P = 0.025$  and cg00298481  $P = 0.035$ ). Given that exercise mediates BDNF action on cognitive plasticity, exercise may be a non-invasive, drug free option in the treatment of PTSD.

**Keywords:** PTSD, DNA methylation, exercise, BDNF

## 1. Introduction

Posttraumatic stress disorder (PTSD) occurs following exposure to a potentially life threatening traumatic event. The trauma can cause significant changes in the brain that influence future responses to stress. PTSD can result from varied traumatic events. Stress after military combat is one of the most common causes but violent assaults, terrorism encounters and serious accidents can all result in the development of PTSD.

Not all individuals that are exposed to a life threatening experience will develop PTSD. Twin studies of Vietnam veterans have confirmed that genetic vulnerability is associated with the disorder [1-3]. Approximately 30% of the variance in liability for PTSD symptoms (including re-experiencing, avoidance and hyper-arousal) was found to be genetic after analysing monozygotic and dizygotic twins [1-3]. More recently, using genome-wide genetic data from over 20 000 individuals, the Psychiatric Genetics Consortium (PGC-PTSD) has identified the genetic risk at 29 % in European American females [4].

A number of genes have been identified that are associated with increased risk of PTSD (reviewed in [5]) including the brain-derived neurotrophic factor gene (*BDNF*). *BDNF* belongs to the neurotrophin family, is expressed in nearly all brain regions and plays a role in development, neuroplasticity, cognition and memory (reviewed in [6]).

A non-synonymous polymorphism (rs6265; Val66Met) in *BDNF* was found to be associated with psychotic symptoms in PTSD [7]. In a meta-analysis we also identified a trend association of *BDNF* with PTSD [8]. The Val66Met polymorphism has been shown to be functionally relevant as it affects *BDNF* protein production [9].

The interaction of genes and environment determines whether an individual is more likely to have resilience to a traumatic event or later develop PTSD. The heritability of PTSD

vulnerability is high [1, 2, 10]; yet much of the susceptibility to PTSD is due to environmental effects. Because environmental processes play an important part in PTSD aetiology, epigenetic studies hold promise for identifying novel candidate genes and pathways. DNA methylation is one form of epigenetic modification and it has been found to play a role in the pathophysiology of PTSD. Methylation studies have identified a number of genes that are implicated in PTSD including *FKBP5*, *COMT* and *NR3C1* (see reviews [11, 12]). Using whole genome methylation analysis we recently identified several genes associated with differential methylation in war veterans including *DOCK2* that reached genome wide significance and was replicated in another cohort [13]. In this study we also identified that differential *BDNF* methylation was associated with increased PTSD symptom severity. A recent report has similarly documented the association between *BDNF* methylation and a diagnosis of PTSD in Korean Vietnam veterans [14].

Exercise is an important environmental factor that also regulates epigenetic mechanisms. It is believed that regular exercise either short or long term, reduces inflammation which can modify epigenetic mechanisms (reviewed in [15]). The impact of exercise on epigenetic modification is also gene specific and not all genes will have a decrease in methylation following exercise [16]. Research has shown that physical activity increases *BDNF* gene and protein expression that activates signalling pathways resulting in enhanced learning and memory [17] [18]. DNA methylation may be a key factor in regulating exercise induced *BDNF* expression.

In the current study we sought to further investigate the role of *BDNF* in PTSD using genetic, epigenetic and gene expression analysis. We also investigated the implications of exercise in mediating the effects of *BDNF*.

## 2. Materials and Methods

### 2.1 Participants

A total of 299 male Vietnam veterans were recruited through Greenslopes Private Hospital and Returned and Services League (RSL) Queensland branch by the Gallipoli Medical Research Foundation (GMRF). For inclusion in the study, participants had to have been deployed to Vietnam during the Vietnam War by the Australian or New Zealand Defence Force. The mean age of the all-male cohort was 68.82 years ( $SD = 4.2$ ) at recruitment. PTSD diagnostic status was determined through structured interviews by psychiatrists. A total of 159 participants received a PTSD diagnosis, and 140 participants did not meet diagnostic criteria despite comparable trauma-exposure.

During medical assessment a blood sample was taken from each participant for DNA and RNA analysis. Genome-wide DNA methylation and expression analysis was performed in a subset of 96 individuals (previously described [13]). Cases were combat-exposed veterans with current PTSD ( $n = 48$ ) and controls were combat-exposed veterans with no current PTSD or previous PTSD diagnosis ( $n = 48$ ). Table 1 shows an overview of the characteristics of the 96 sample cohort by diagnostic status.

Each participant gave written informed consent before commencement of their involvement in the project. Ethics approval was obtained from the Human Research Ethics Committee of the Queensland University of Technology and Greenslopes Research and Ethics Committee. This study was carried out in accordance with The Code of Ethics of the World Medical Association (Declaration of Helsinki).

### 2.2 Clinical Assessments

Severity of PTSD symptoms was assessed by clinical psychologists using the Clinician-Administered PTSD Scale for DSM-5 (CAPS-5). The Depression Anxiety Stress Scale 21



(DASS-21) scale was used to measure stress, depression and anxiety. The PHQ (patient health questionnaire 9) was also used to screen for the presence and severity of depression. The Connor-Davidson Resilience Scale (CD RISC) was used to measure resilience. Alcohol use disorder was assessed using World Health Organisation's Alcohol Use Disorders Identification Test AUDIT[19]. The AUDIT is a 10-item questionnaire designed to identify hazardous drinkers, harmful drinkers, and individuals with severe alcohol use disorder[19]. Medical history was collected for each participant by medical officers using a semi-structured interview. Structured clinical history was also collected including demographic information, smoking, diet, exercise, life-time history of alcohol consumption, past and current illnesses, medications and family medical history. The Montreal cognitive assessment (MOCA) score was used to assess mild cognitive impairment.

### 2.3 *Genotyping, Methylation and Expression Analysis*

Blood samples were sent to the Australian Genome Research Facility (AGRF) for DNA and RNA extraction, genotyping, DNA methylation and gene expression analysis. Genomic DNA was extracted from a 2 ml blood sample, using MACHERY-NAGEL NucleoSpin L (MACHERY-NAGEL GmbH & Co. KG, Dueren, NRW, Germany). Quality of the DNA was assessed through resolution on a 0.8% agarose gel at 130 V for 60 minutes. Samples were normalised to 200 ng of DNA in 4 µl.

Genotyping was performed using the Illumina PsychArray-24 BeadChip scanned with the Illumina iScan systems. Data were exported and quality checks were performed using GenomeStudio v2011.1 (Illumina, San Diego, CA, USA) with Genotyping Module 1.9.4 software using the InfiniumPsychArray-24v1-1\_A1 manifest and cluster files to call the genotypes. After quality control, the genotype call rate was 99.59% across the remaining

variants. Eleven samples were performed in duplicate for quality assurance with 100 % reproducibility for the *BDNF* SNP of interest rs6265 (Val66Met).

For DNA methylation, samples were bisulphite converted with Zymo EZ DNA Methylation kit as previously described [20, 21]. The Infinium platform assays more than 850,000 CpG sites, encompassing 99% of RefSeq genes. It covers 96% of CpG islands with multiple sites in the island, the shores (within 2 kb from CpG islands) and the shelves (> 2 kb from CpG islands). It also covers CpG sites outside of CpG islands and DNase hypersensitive sites as well as incorporating miRNA promoter regions.

RNA was extracted from a 2.5 ml PaxGene (PreAnalytiX GmbH, Switzerland) blood sample, using MACHERY-NAGEL NucleoSpin RNA Blood kit (MACHERY-NAGEL GmbH & Co. KG, Dueren, NRW, Germany). The quality and quantity of total RNA was analysed on the Agilent Bioanalyser 2100 using the NanoChip protocol. A total of 500ng of total RNA was then prepared for hybridisation to the Illumina HumanHT 12 Expression Beadchip (array content HumanHT-12\_V4\_0\_R2\_15002873\_B) as outlined in the Illumina manual and scanned in the Illumina iScan Reader.

## 2.4 Statistical analysis

Raw  $\beta$  values from EPIC Illumina arrays were exported into R (<https://www.r-project.org/>) for statistical analysis. For quality control (QC), pairs of technical and biological replicates were included to test and confirm the robustness of the microarray procedures and all replicate correlations were high ( $r > 0.97$ ), indicating high reproducibility. Samples with probe detection call rates < 95% as well as those with an average intensity value of either <50% of the experiment-wide sample mean or < 2000 arbitrary units (AU) were excluded from further analysis. The level of methylation was determined by calculating a “ $\beta$  value” -

the ratio of the fluorescent signals for the methylated versus unmethylated sites. Intensity read outs, normalisation and methylation  $\beta$  values calculations were performed using the minfi Bioconductor R package version 1.10.2. For methylation analysis, IDAT files were loaded into the R (2.15) environment. The arrays were then background and control normalised using the minfi package [22]. Subset-quantile Within-Array Normalisation (SWAN) was used to remove technical differences between Infinium I and Infinium II probes available in the minfi package [23, 24]. The methylation status for each probe was recorded as a  $\beta$ -value that ranged between 0 and 1, where values close to 1 represent high levels of methylation and where values close to 0 represent low levels of methylation. After normalisation, seven *BDNF* CpG sites remained for analysis. The location of the CpG sites can be found in Supplementary Table 1.

Gene expression data were transformed and normalised using the variance stabilising normalisation [25]. Probes passing the filter criteria of Illumina probe detection p-value of  $< 0.05$  in at least 5% of the samples were used for subsequent analysis.

Genotypes were obtained from the Illumina Psych array for all the 96 veterans with DNA methylation data. Initial quality control was performed in PLINK 1.9 [26, 27] - SNPs with  $>1\%$  missing values, a Hardy-Weinberg equilibrium  $P$ -Value  $< 0.0001$  and a minor allele frequency of  $< 5\%$  were excluded, resulting in 302,607 genotyped SNPs. The genotypes were recoded as vcf files before uploading to the Michigan Imputation Server (<https://imputationserver.sph.umich.edu/start.html#!pages/home>) using SHAPEIT to phase haplotypes, and Minimac with the most recent 1000 Genomes reference panel (phase 3, version 5). Imputed genotypes were then filtered and recoded in PLINK removing samples with  $>5\%$  missing values, and multi-allelic and monomorphic SNPs, SNPs  $>1\%$  missing values, a Hardy-Weinberg equilibrium  $P < 0.0001$ , a minor allele frequency of  $< 5\%$ .

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### 3. Results

#### 3.1 PTSD genetic association with Val66Met (rs6265) genotype

A= Met, G= Val

No significant difference in genotype counts was observed between PTSD cases and controls using Pearson Chi-Squared ( $P = 0.623$ ). PTSD group genotype counts: AA = 3; AG = 14; GG = 31. Non PTSD group genotype counts: AA = 4; AG = 10; GG = 34.

Analysis of Variance (ANOVA) was used to test the association of clinical measures with genotype. No significant differences were observed between genotype and CAPS-5 total symptom severity, AUDIT total or DASS 21 anxiety, depression and stress total scores. A Trend association was observed between genotype and CDRISC total score ( $P = 0.074$ ). Post-hoc analysis revealed a dominant model and therefore genotypes were combined for further analysis. When the AG and GG genotypes were combined and compared against the AA genotype a significant association was found between genotype and CDRISC ( $P = 0.038$ ). The AA genotype (Met/Met) was associated with increased CDRISC score. Genotype associations did not survive multiple corrections.

#### 3.2 Association of *BDNF* mRNA with PTSD

Using ANOVA no significant differences in *BDNF* mRNA expression were observed between PTSD cases and controls for all four transcript probes on the Illumina array (ILMN\_1719820, ILMN\_1761910, ILMN\_1712712, ILMN\_1793988).

#### 3.3 Association of *BDNF* mRNA expression with genotype

*BDNF* transcript ILMN\_1712712 (contains the longest transcript 1 also known as IX) was significantly associated with differences in *BDNF* mRNA expression between genotypes ( $P =$

0.024) with the AA (Met/Met) associated with decreased mRNA levels. Using Benjamini Hochberg this association did not survive multiple correction.

### 3.4 Association of *BDNF* methylation with PTSD status and PTSD severity

After normalisation seven *BDNF* methylation sites were analysed for association with PTSD using ANOVA. Supplementary Table 1 describes the seven sites and location across the genome. Whole blood cell counts were estimated from the DNA methylation data implemented in the minfi package in R [22]. The following cell types were estimated: CD8, CD4, NK, B cells, monocytes and granulocytes. Monocytes were significantly different between the PTSD and non-PTSD group ( $P = 0.003$ ). Other cell types were not significantly different between groups. Since differences in monocytes were found between the PTSD and non-PTSD group, cell counts were selected as covariates in the regression analyses. Methylation levels were significantly different between PTSD cases and controls for ccg01546433 ( $P = 0.004835$ ), cg24650785 ( $P = 0.000259$ ) and cg002298481 ( $P = 0.000672$ ) (Figure 1) with decreased methylation associated with PTSD.

For cg04481212, cg05733135, cg06684850 and cg15710245 methylation levels were not significantly different between groups.

Using Benjamini Hochberg, cg01546433, cg24650785 and cg002298481 all survived multiple correction.

PTSD symptom severity was correlated with the three significant *BDNF* CpGs and a weak negative relationship was found with cg01546433 ( $r = -0.276$ ;  $P = 0.006$ ), cg24650785 ( $r = -0.327$ ;  $P = 0.001$ ) and cg002298481 ( $r = -0.268$ ;  $P = 0.008$ ). At each of three sites, decreased methylation was associated with increased severity levels. Using Benjamini Hochberg, cg01546433, cg24650785 and cg002298481 all survived multiple correction.

### 3.5 *Correlation of BDNF Methylation with Expression*

No significant correlations were observed between BDNF expression and methylation.

### 3.6 *Association of BDNF methylation with genotype*

No significant differences were found between genotype and methylation levels for any of the seven *BDNF* CpG sites using ANOVA.

### 3.7 *Association of BDNF methylation and mRNA with exercise*

Participants were asked if they exercised regularly including walking, swimming, running, bike riding, team sports or other physical activity. Active exercise was defined as regular exercise that was completed three or more times /week. Differential *BDNF* methylation was found associated with exercise at cg04481212 ( $P = 0.005$ ), cg01546433 ( $P = 0.025$ ) and cg00298481 ( $P = 0.035$ ). All of these CpG sites survived multiple correction using Benjamini Hochberg. Active exercise was associated with lower methylation levels at all three CpG sites.

Exercise was not associated with differential *BDNF* mRNA expression after analysing all four transcripts.

### 3.8 *BDNF methylation to predict PTSD*

Binary logistic regression was used to predict PTSD using methylation levels. Age, BMI, cell counts (CD8T, CD4T, Natural Killer cells (NK), B cells, monocytes and granulocytes), alcohol history and psychotropic medication were also used as covariates in the model since they were significantly associated with PTSD in our analyses or are known predictors of PTSD. Only two CpG methylation sites (cg 01546433 and cg 24650785) were used in the model as the others were significantly correlated with each other. The Hosmer and

Lemeshow goodness-of-fit statistic was 2.139 ( $P = 0.976$ ). The overall percentage of correct classification was 91.7 %. The correct classification for no PTSD was 88.1 % and for PTSD was 95.2 %.

#### 4 Discussion

In this Vietnam veteran study we found that the *BDNF* Val66Met was associated with differential *BDNF* expression with the Met/Met genotype associated with decreased expression in peripheral tissue. This result is consistent with other studies that found the Met allele is associated with reduced gene expression and lower hippocampal volume [9, 28].

Although studies have found reduced *BDNF* mRNA expression in participants with depression [29, 30] we did not find evidence of reduced *BDNF* in our PTSD cohort.

Medication may also influence *BDNF* expression as a study has shown that participants with depression who are medication-free have reduced peripheral *BDNF* mRNA levels and that 12 weeks of antidepressant treatment is able to reverse this effect [29].

A diagnosis of PTSD was associated with differential *BDNF* methylation in this Vietnam veteran cohort. Lower *BDNF* methylation at three CpG sites was found in veterans with PTSD. We also observed an association between PTSD severity and *BDNF* methylation at these same CpG sites. Decreased *BDNF* methylation was associated with increased PTSD symptom severity. A recent study found differential *BDNF* methylation associated with PTSD in Vietnam veterans of Korean origin [14]. The Korean study found increased *BDNF* methylation was associated with PTSD which is discordant with our findings. Decreased *BDNF* methylation observed in our PTSD cohort may be the result of 70% of PTSD participants currently taking psychotropic medication. Anti-depressant medication and psychotherapy is known to decrease *BDNF* methylation [31] [32]. A further study found patients with bipolar disorder had decreased *BDNF* methylation after psychotherapy but the



effect was only evident in responders [33]. Non-responders were found to have increased *BDNF* methylation. The Perroud et al. study confirmed that DNA methylation is dynamic and can be altered not only by medication but also therapy. Control participants in the current study were also trauma exposed which may have altered *BDNF* methylation profiles. It is possible that combat exposure alone alters *BDNF* methylation and the diagnosed PTSD cohort are seeking treatment which is resulting in decreased *BDNF* methylation compared to our combat exposed controls. A review by Zheleznyakova et al. also highlights other studies that show decreased *BDNF* methylation is associated with psychiatric disorders [34]

Although we did not find any difference in exercise habits between the participants with and without PTSD we did find that differential *BDNF* methylation was associated with exercise. In the total cohort of combat exposed veterans, active exercise was associated with lower *BDNF* methylation. This result has important implications for the effects of exercise on *BDNF* modulation. An early study in an animal model found that exercise enhanced learning and memory and was associated with increased *BDNF* mRNA [18]. Furthermore, in the rat, exercise has been found to decrease *BDNF* methylation and increase *BDNF* mRNA [35]. Recent reviews have outlined the importance of exercise in treating mental illness [36].

*BDNF* is likely to play a role in other psychiatric disorders as we have previously identified association between *BDNF* expression and methylation levels in schizophrenia [37]. The PGC-PTSD also recently confirmed strong evidence of overlapping genetic risk between schizophrenia and PTSD in over 20 000 individuals [4].

Limitations of the study include (i) the use of an all-male cohort; future studies should investigate *BDNF* in combat exposed females to identify gender specific epigenetic markers. (ii) the samples included an ageing population and therefore exposure to environmental factors modifying DNA methylation is increased; a replication using a younger veteran

cohort is recommended. (iii) this study was performed in peripheral tissue and not post-mortem brain samples. However a study has identified that peripheral *BDNF* methylation correlates with brain *BDNF* methylation [38]. (iv) our sample size was relatively small and should be replicated in a larger cohort. (v) Since methylation levels were not measured directly after trauma exposure it is difficult to determine if differential methylation is directly associated with PTSD status or influenced by various treatments such as antidepressants.

There are a number of strengths of the study (i) the sample was homogeneous including age and gender matched and clinically well characterised. (ii) all participants had combat exposure including the controls allowing control for the type of trauma and the experience of military life.

**5. Conclusions:** In summary, *BDNF* has strong potential as a therapeutic target as several studies have shown that antidepressant medication, electroconvulsive therapy and exercise are able to modulate *BDNF* expression. Given that exercise mediates *BDNF* action on cognitive plasticity, exercise may be an important treatment component for PTSD.

## Supplementary Material

Supplementary Table 1.

## Abbreviations

PTSD: Posttraumatic stress disorder; CpG: cytosine-phosphate-guanine dinucleotide; PGC-PTSD: Psychiatric Genetics Consortium; *NR3C1*::glucocorticoid receptor gene; *FKBP5*: FK506 Binding Protein 5 gene; *COMT*: Catechol-O-methyltransferase gene; *DOCK2*: Dedicator Of Cytokinesis 2 gene; GMRF: Gallipoli Medical Research Foundation; QUT: Queensland University of Technology; GPH: Greenslopes Private Hospital; DSM-IV: Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition; CAPS-5: Clinician Administered PTSD Scale for DSM-5; MINI : Mini International Neuropsychiatric Interview DSM IV; DASS-21: Depression Anxiety Stress Scale 21; PHQ-9: Patient Health Questionnaire PHQ-9; PRIME-MD : Primary Care Evaluation of Mental Disorders; MDD: Major Depressive Disorder; CD RISC: Connor-Davidson Resilience Scale; AGRF: Australian Genome Research Facility AGRF; SE: standard error; IHBI: Institute of Health and Biomedical Innovation;

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## Conflict of Interest

None

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## References

- [1] Koenen KC, Harley R, Lyons MJ, Wolfe J, Simpson JC, Goldberg J, Eisen SA, Tsuang M. A twin registry study of familial and individual risk factors for trauma exposure and posttraumatic stress disorder. *The Journal of nervous and mental disease*. 2002;190:209-18.
- [2] True WR, Rice J, Eisen SA, Heath AC, Goldberg J, Lyons MJ, Nowak J. A twin study of genetic and environmental contributions to liability for posttraumatic stress symptoms. *Arch Gen Psychiatry*. 1993;50:257-64.
- [3] Lyons MJ, Goldberg J, Eisen SA, True W, Tsuang MT, Meyer JM, Henderson WG. Do genes influence exposure to trauma? A twin study of combat. *Am J Med Genet*. 1993;48:22-7.
- [4] Duncan LE, Ratanatharathorn A, Aiello AE, Almli LM, Amstadter AB, Ashley-Koch AE, Baker DG, Beckham JC, Bierut LJ, Bisson J, Bradley B, Chen CY, Dalvie S, Farrer LA, Galea S, Garrett ME, Gelernter JE, Guffanti G, Hauser MA, Johnson EO, Kessler RC, Kimbrel NA, King A, Koen N, Kranzler HR, Logue MW, Maihofer AX, Martin AR, Miller MW, Morey RA, Nugent NR, Rice JP, Ripke S, Roberts AL, Saccone NL, Smoller JW, Stein DJ, Stein MB, Sumner JA, Uddin M, Ursano RJ, Wildman DE, Yehuda R, Zhao H, Daly MJ, Liberzon I, Ressler KJ, Nievergelt CM, Koenen KC. Largest GWAS of PTSD (N=20 070) yields genetic overlap with schizophrenia and sex differences in heritability. *Mol Psychiatry*. 2017.
- [5] Yehuda R, Koenen KC, Galea S, Flory JD. The role of genes in defining a molecular biology of PTSD. *Dis Markers*. 2011;30:67-76.
- [6] Kowianski P, Lietzau G, Czuba E, Waskow M, Steliga A, Morys J. BDNF: A Key Factor with Multipotent Impact on Brain Signaling and Synaptic Plasticity. *Cellular and molecular neurobiology*. 2017.
- [7] Pivac N, Kozaric-Kovacic D, Grubisic-Ilic M, Nedic G, Rakos I, Nikolac M, Blazev M, Muck-Seler D. The association between brain-derived neurotrophic factor Val66Met variants

- and psychotic symptoms in posttraumatic stress disorder. *World J Biol Psychiatry*. 2012;13:306-11.
- [8] Bruenig D, Lurie J, Morris CP, Harvey W, Lawford B, Young RM, Voisey J. A Case-Control Study and Meta-Analysis Reveal BDNF Val66Met Is a Possible Risk Factor for PTSD. *Neural Plast*. 2016;2016:6979435.
- [9] Egan MF, Kojima M, Callicott JH, Goldberg TE, Kolachana BS, Bertolino A, Zaitsev E, Gold B, Goldman D, Dean M, Lu B, Weinberger DR. The BDNF val66met polymorphism affects activity-dependent secretion of BDNF and human memory and hippocampal function. *Cell*. 2003;112:257-69.
- [10] Sartor CE, Grant JD, Lynskey MT, McCutcheon VV, Waldron M, Statham DJ, Bucholz KK, Madden PA, Heath AC, Martin NG, Nelson EC. Common heritable contributions to low-risk trauma, high-risk trauma, posttraumatic stress disorder, and major depression. *Arch Gen Psychiatry*. 2012;69:293-9.
- [11] Ryan J, Chaudieu I, Ancelin ML, Saffery R. Biological underpinnings of trauma and post-traumatic stress disorder: focusing on genetics and epigenetics. *Epigenomics*. 2016;8:1553-69.
- [12] Voisey J, Young RM, Lawford BR, Morris CP. Progress towards understanding the genetics of posttraumatic stress disorder. *J Anxiety Disord*. 2014;28:873-83.
- [13] Mehta D, Bruenig D, Carrillo-Roa T, Lawford B, Harvey W, Morris CP, Smith AK, Binder EB, Young RM, Voisey J. Genomewide DNA methylation analysis in combat veterans reveals a novel locus for PTSD. *Acta Psychiatr Scand*. 2017;136:493-505.
- [14] Kim TY, Kim SJ, Chung HG, Choi JH, Kim SH, Kang JI. Epigenetic alterations of the BDNF gene in combat-related post-traumatic stress disorder. *Acta Psychiatr Scand*. 2017;135:170-9.
- [15] Horsburgh S, Robson-Ansley P, Adams R, Smith C. Exercise and inflammation-related epigenetic modifications: focus on DNA methylation. *Exerc Immunol Rev*. 2015;21:26-41.
- [16] Voisin S, Eynon N, Yan X, Bishop DJ. Exercise training and DNA methylation in humans. *Acta Physiol (Oxf)*. 2015;213:39-59.
- [17] Sleiman SF, Henry J, Al-Haddad R, El Hayek L, Abou Haidar E, Stringer T, Ulja D, Karuppagounder SS, Holson EB, Ratan RR, Ninan I, Chao MV. Exercise promotes the expression of brain derived neurotrophic factor (BDNF) through the action of the ketone body beta-hydroxybutyrate. *Elife*. 2016;5.
- [18] Vaynman S, Ying Z, Gomez-Pinilla F. Hippocampal BDNF mediates the efficacy of exercise on synaptic plasticity and cognition. *Eur J Neurosci*. 2004;20:2580-90.
- [19] Saunders JB, Aasland OG, Babor TF, De la Fuente JR, Grant M. Development of the alcohol use disorders identification test (AUDIT): WHO collaborative project on early detection of persons with harmful alcohol consumption-II. *Addiction*. 1993;88:791-804.
- [20] Wockner LF, Noble EP, Lawford BR, Young RM, Morris CP, Whitehall VL, Voisey J. Genome-wide DNA methylation analysis of human brain tissue from schizophrenia patients. *Transl Psychiatry*. 2014;4:e339.
- [21] Mehta D, Klengel T, Conneely KN, Smith AK, Altmann A, Pace TW, Rex-Haffner M, Loeschner A, Gonik M, Mercer KB, Bradley B, Muller-Myhsok B, Ressler KJ, Binder EB. Childhood maltreatment is associated with distinct genomic and epigenetic profiles in posttraumatic stress disorder. *Proc Natl Acad Sci U S A*. 2013;110:8302-7.
- [22] Aryee MJ, Jaffe AE, Corrada-Bravo H, Ladd-Acosta C, Feinberg AP, Hansen KD, Irizarry RA. Minfi: a flexible and comprehensive Bioconductor package for the analysis of Infinium DNA methylation microarrays. *Bioinformatics*. 2014;30:1363-9.
- [23] Pidsley R, CC YW, Volta M, Lunnon K, Mill J, Schalkwyk LC. A data-driven approach to preprocessing Illumina 450K methylation array data. *BMC Genomics*. 2013;14:293.

- [24] Maksimovic J, Gordon L, Oshlack A. SWAN: Subset-quantile within array normalization for illumina infinium HumanMethylation450 BeadChips. *Genome Biol.* 2012;13:R44.
- [25] Huber W, von Heydebreck A, Sultmann H, Poustka A, Vingron M. Variance stabilization applied to microarray data calibration and to the quantification of differential expression. *Bioinformatics.* 2002;18 Suppl 1:S96-104.
- [26] Chang CC, Chow CC, Tellier LC, Vattikuti S, Purcell SM, Lee JJ. Second-generation PLINK: rising to the challenge of larger and richer datasets. *Gigascience.* 2015;4:7.
- [27] Purcell S, Neale B, Todd-Brown K, Thomas L, Ferreira MA, Bender D, Maller J, Sklar P, de Bakker PI, Daly MJ, Sham PC. PLINK: a tool set for whole-genome association and population-based linkage analyses. *Am J Hum Genet.* 2007;81:559-75.
- [28] Hariri AR, Goldberg TE, Mattay VS, Kolachana BS, Callicott JH, Egan MF, Weinberger DR. Brain-derived neurotrophic factor val66met polymorphism affects human memory-related hippocampal activity and predicts memory performance. *J Neurosci.* 2003;23:6690-4.
- [29] Cattaneo A, Bocchio-Chiavetto L, Zanardini R, Milanese E, Placentino A, Gennarelli M. Reduced peripheral brain-derived neurotrophic factor mRNA levels are normalized by antidepressant treatment. *Int J Neuropsychopharmacol.* 2010;13:103-8.
- [30] Cattaneo A, Gennarelli M, Uher R, Breen G, Farmer A, Aitchison KJ, Craig IW, Anacker C, Zunsztain PA, McGuffin P, Pariante CM. Candidate genes expression profile associated with antidepressants response in the GENDEP study: differentiating between baseline 'predictors' and longitudinal 'targets'. *Neuropsychopharmacology.* 2013;38:377-85.
- [31] Tadic A, Muller-Engling L, Schlicht KF, Kotsiari A, Dreimuller N, Kleimann A, Bleich S, Lieb K, Frieeling H. Methylation of the promoter of brain-derived neurotrophic factor exon IV and antidepressant response in major depression. *Mol Psychiatry.* 2014;19:281-3.
- [32] Yehuda R, Daskalakis NP, Desarnaud F, Makotkine I, Lehrner AL, Koch E, Flory JD, Buxbaum JD, Meaney MJ, Bierer LM. Epigenetic Biomarkers as Predictors and Correlates of Symptom Improvement Following Psychotherapy in Combat Veterans with PTSD. *Frontiers in psychiatry.* 2013;4:118.
- [33] Perroud N, Salzmann A, Prada P, Nicastro R, Hoeppli ME, Furrer S, Ardu S, Krejci I, Karege F, Malafosse A. Response to psychotherapy in borderline personality disorder and methylation status of the BDNF gene. *Translational psychiatry.* 2013;3:e207.
- [34] Zheleznyakova GY, Cao H, Schioth HB. BDNF DNA methylation changes as a biomarker of psychiatric disorders: literature review and open access database analysis. *Behav Brain Funct.* 2016;12:17.
- [35] Gomez-Pinilla F, Zhuang Y, Feng J, Ying Z, Fan G. Exercise impacts brain-derived neurotrophic factor plasticity by engaging mechanisms of epigenetic regulation. *Eur J Neurosci.* 2011;33:383-90.
- [36] Lederman O, Suetani S, Stanton R, Chapman J, Korman N, Rosenbaum S, Ward PB, Siskind D. Embedding exercise interventions as routine mental health care: implementation strategies in residential, inpatient and community settings. *Australas Psychiatry.* 2017;25:451-5.
- [37] Cheah SY, McLeay R, Wockner LF, Lawford BR, Young RM, Morris CP, Voisey J. Expression and methylation of BDNF in the human brain in schizophrenia. *World J Biol Psychiatry.* 2017;18:392-400.
- [38] Stenz L, Zewdie S, Laforge-Escarra T, Prados J, La Harpe R, Dayer A, Paoloni-Giacobino A, Perroud N, Aubry JM. BDNF promoter I methylation correlates between post-mortem human peripheral and brain tissues. *Neurosci Res.* 2015;91:1-7.

**Table 1. Demographic and clinical comparisons between the PTSD and non PTSD group.**

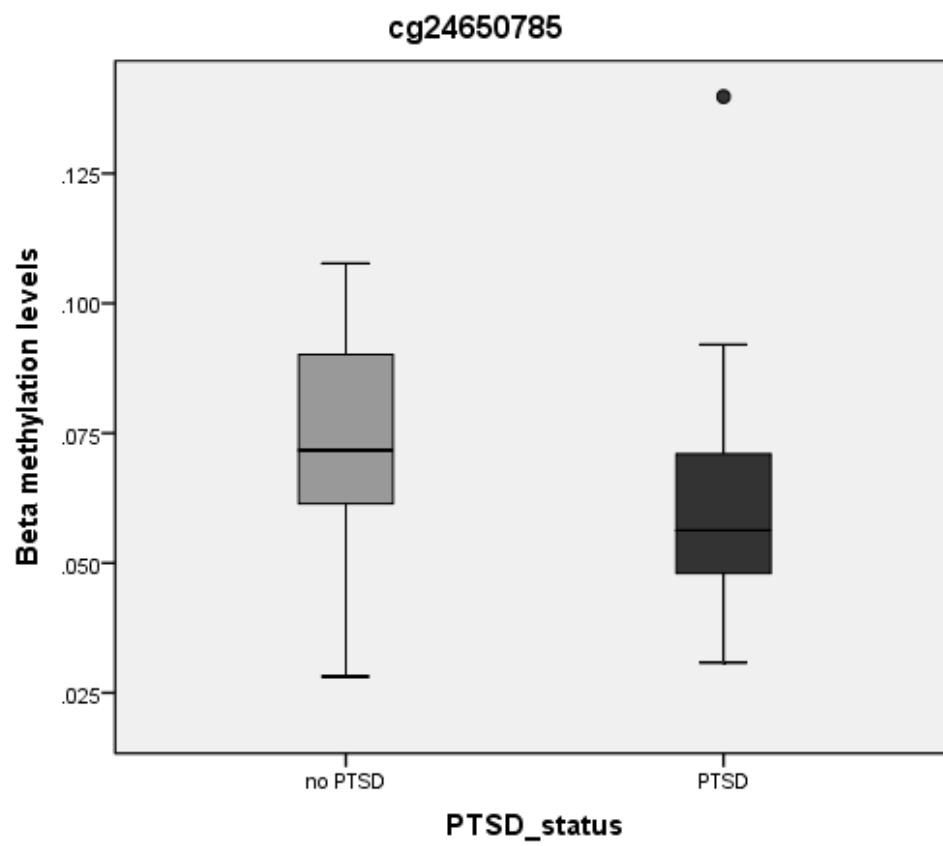
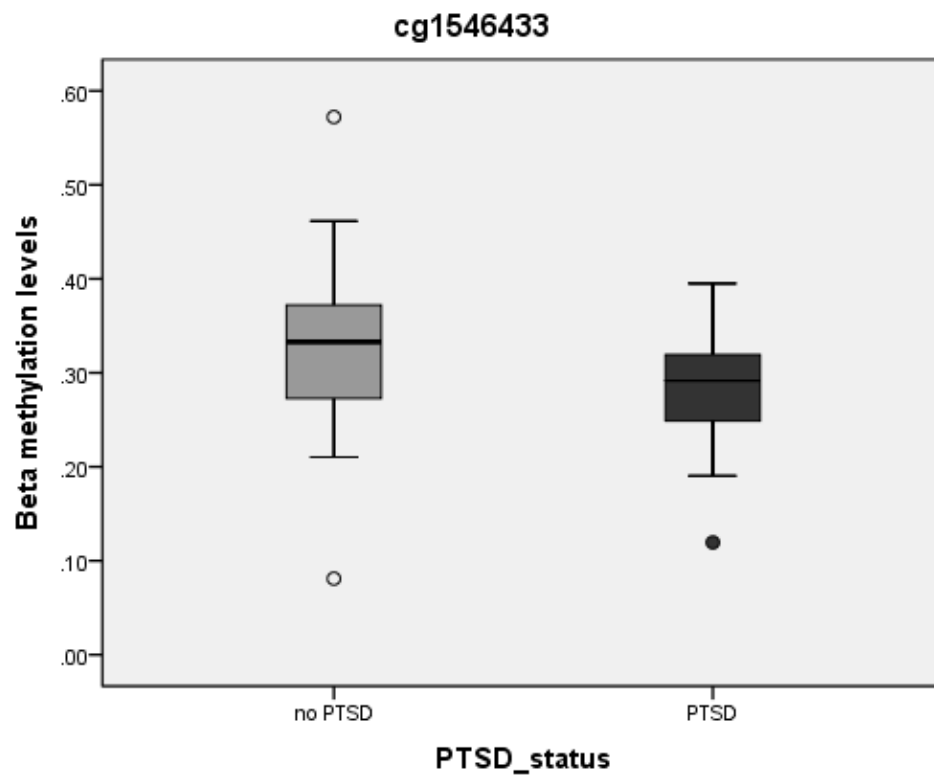
| Characteristic                                           | PTSD N=48    | No PTSD N = 48 | P-value |
|----------------------------------------------------------|--------------|----------------|---------|
| Age: mean years (SE)                                     | 67.94 (0.61) | 69.40 (0.64)   | 0.10    |
| Marital status: % married                                | 87%          | 85%            | 0.77    |
| BMI: mean (SE)                                           | 29.93 (0.74) | 29.13 (0.54)   | 0.38    |
| Psychotropic medication % yes                            | 71%          | 4.8%           | < 0.001 |
| Antidepressant medication                                | 69%          | 2.4%           | < 0.001 |
| Antipsychotic medication                                 | 7%           | 0%             | 0.120   |
| benzodiazepines                                          | 26%          | 2.4%           | 0.001   |
| Alcohol history: % moderate to heavy intake <sup>a</sup> | 92 %         | 56 %           | < 0.001 |
| Exercise <sup>b</sup> (% active)                         | 77%          | 75 %           | 0.81    |
| Total Symptom Severity (CAPS) mean (SE)                  | 25.92 (1.25) | 1.19 (0.31)    | < 0.001 |
| CDRISC25 total resilience score mean (SE)                | 64.90 (2.15) | 80.85 (1.69)   | < 0.001 |
| DASS21 Depression score mean (SE)                        | 5.40 (0.78)  | 1.46 (0.24)    | < 0.001 |
| DASS21 Anxiety score mean (SE)                           | 8.17 ( 0.68) | 0.81 (0.18)    | < 0.001 |
| DASS21 Stress score mean (SE)                            | 12.38 (0.71) | 3.81 (0.46)    | < 0.001 |

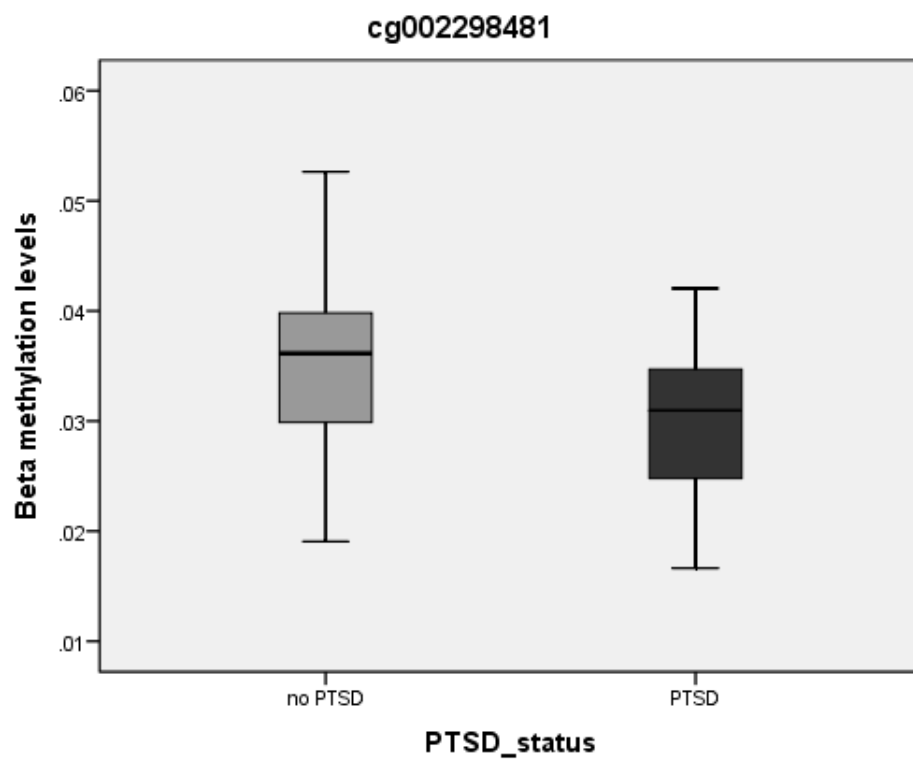
|                                                            |              |              |         |
|------------------------------------------------------------|--------------|--------------|---------|
| PHQ (patient health questionnaire 9) total score mean (SE) | 11.23 (0.84) | 2.00 (0.35)  | < 0.001 |
| MoCA (Montreal cognitive assessment) total score mean (SE) | 25.85 (0.42) | 26.42 (0.33) | 0.30    |

<sup>a</sup>alcohol history included combined moderate (heavy drinking for > 3 years) and heavy drinking (lifetime) versus no or low (no regular drinking).

<sup>b</sup>Active exercise was measured as regular exercise > 3 times/week.







**Figure 1.** Boxplots of differential *BDNF* DNA methylation in veterans with and without PTSD.

## Abbreviations

PTSD: Posttraumatic stress disorder; CpG: cytosine-phosphate-guanine dinucleotide; PGC-

PTSD: Psychiatric Genetics Consortium; *NR3C1*::glucocorticoid receptor gene; *FKBP5*:

FK506 Binding Protein 5 gene; *COMT*: Catechol-O-methyltransferase gene; *DOCK2*:

Dedicator Of Cytokinesis 2 gene; GMRF: Gallipoli Medical Research Foundation; QUT:

Queensland University of Technology; GPH: Greenslopes Private Hospital; DSM-IV:

Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition; CAPS-5: Clinician

Administered PTSD Scale for DSM-5; MINI : Mini International Neuropsychiatric Interview

DSM IV; DASS-21: Depression Anxiety Stress Scale 21; PHQ-9: Patient Health

Questionnaire PHQ-9; PRIME-MD : Primary Care Evaluation of Mental Disorders; MDD:

Major Depressive Disorder; CD RISC: Connor-Davidson Resilience Scale; AGRF:

Australian Genome Research Facility AGRF; SE: standard error; IHBI: Institute of Health and Biomedical Innovation.

**Highlights**

Three *BDNF* CpG sites were significantly associated with PTSD in combat exposed Vietnam veterans.

Differential *BDNF* methylation was associated with exercise, with active exercise associated with lower methylation levels at three *BDNF* CpG sites.

No association between *BDNF* mRNA and PTSD was identified.